1 Question (12 points)

$$
\begin{aligned}
E &\rightarrow E + T \\
E &\rightarrow T \\
T &\rightarrow T * F \\
T &\rightarrow F \\
F &\rightarrow (E) \\
F &\rightarrow a
\end{aligned}
$$

For the given CFG above.

a. What is V (Set of variables)?

b. What is Σ (Set of terminals)?

c. What is the start variable?

d. Give two distinct parse trees for the expression a + a * a

2 Question (20 points)

Give PDA's that generate the following languages. The alphabet is $\{0, 1\}$.

- A = $\{$w | w contains at least two 0s and two 1s$\}$
- B = $\{$w | the length of w is even and its middle symbols are 00 or 11$\}$

3 Question (18 points)

Show that the class of context-free languages is closed under the regular operations,

a. Union

b. Concatenation

c. Star

(Hint: Add an extra variable to combine the two CFG's of the languages)

4 Question (16 points)

Convert the following grammar into CNF.

$$\begin{aligned} S &\rightarrow A1A \\ A &\rightarrow 0B0 \\ A &\rightarrow \epsilon \\ B &\rightarrow A \\ B &\rightarrow 10 \end{aligned}$$

5 Question (16 points)

Give context-free grammars that generate the following languages. The alphabet is $\{0, 1\}$.

- A = {w | w contains at least two 0s and two 1s}
- B = {w | the length of w is even and its middle symbols are 00 or 11}

6 Question (18 points)

Are the following languages context-free? Prove your answer.

- A = $\{a^n b^m c^m d^n | n, m \geq 0\}$
- B = $\{a^n b^m c^n d^m | n, m \geq 0\}$